Extended File Allocation Table

WHAT IS THIS FILESYSTEM?

exFAT is a modern FAT-family format designed for large removable volumes and files. This Infiltrator module provides 64-bit file offset and size operations.

How the disk is organised

Simplified illustration; not a sector-for-sector specification.

What this implementation provides

- Independent exFAT.so with versioned V3/V4/V5 interfaces
- 64-bit file size and offset handling
- Read/write/format media tests against independent exfatprogs images
- Rejects foreign FAT32 media rather than interpreting them as exFAT

One installed module and one manual

```
/System/Filesystems/exFAT.so
/System/Resources/Documentation/Filesystem-exFAT/
exFAT.pdf
```

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How to use the installed module

The exFAT.so driver is a shared library. It is loaded by a compatible filesystem consumer through its published versioned interfaces; it is not a command to run directly.

Installation and verification

Portable package: install the released Debian .deb.

This release publishes a portable .deb, not a native .run.

The Debian package owns exactly one .so and this PDF.

Build records and checksums live in the Debian package metadata.
Full source, tests and engineering notes remain in GitHub.

Current limits and boundaries

- Directory moves remain deliberately refused until ancestry checks are qualified.
- Boot and Defragmenter integration are not yet complete.
- The published exfat.ko is separately qualified for Linux 6.12.107 only.

Operational model

```
Media -> host I/O callbacks -> filesystem .so -> caller
```

The host supplies target selection and protected byte I/O. The filesystem driver interprets that filesystem’s on-disk structures. Ownership, verification and all destructive actions remain the responsibility of the calling application.

This is a user guide, not a claim of untested kernel or firmware support.

Build and installation

Filesystem-exFAT

Release 0.1.6

Installed files

```
/System/Filesystems/exFAT.so

/System/Resources/Documentation/Filesystem-exFAT/exFAT.pdf

/System/Kernel/Filesystems/exfat.ko
```

Debian package

Install the prebuilt files with dpkg -i followed by the Debian filename.

Manual RUN

Run bash followed by the native installer filename.

The installer compiles the shared object and generates this manual locally.

It also compiles the Linux adapter against the running kernel headers.

Use --kernel-build DIRECTORY to select another prepared kernel build.

The manual installer replaces all three files.

InfiltratorOS installation

The OS installer builds a missing or older shared object.

It verifies the manual independently and repairs it from its matching release.

The kernel module is stored for manual use under its single filename.

The package metadata records the source revision and SHA-256 checksums.